

Task 1. ML for Finance. Argimiro Arratia. BSE- 2026

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- Register your team in sheet HW1 of the spreadsheet ML4Finance26_HWteams in the drive.
- Do not hand in R/Python output; program scripts are hand in separately from the report; write report in LaTeX (preferably) or Word (and convert either to PDF); it is not necessary to keep all decimal digits in reported results.
- Put team members names in the report. Teams should be of at most 3 students (and at least 2).
- Always comment your results (as a Data Scientist would do!).
- **Submit your report in PDF format plus R/Python code and extra data (if any, not given out by me), all in a .zip folder via the Classroom with label: “BSE_Task1-Team - members names -”. Only one member of team makes the submission for all. Failing to submit as requested will be penalised –10%**

1. (35%) **Global asset causal network and volatility spillover.** Consider the following 10 ETFs and assets: SPY (S&P 500); XLK (Technology Sector); XLF (Financial Sector); XLE (Energy Sector); GLD (Gold); SLV (Silver); GSG (Broad Commodities); BTC-USD (Bitcoin); ETH-USD (Ethereum); TLT (20+ Year Treasury Bond). The Python script `HW1_getassets.ipynb` (or R script helper) will help you retrieve data (optionally) and organise it for the rest of this exercise. For the epoch assigned to your team in the Google spreadsheet ML4Finance26_HWteams/HW1, must do:

A causality analysis of the **returns** time series among these 10 assets, and a causality analysis of the volatilities series of these assets (this is known as *volatility spill-over*). Estimate volatility using EWMA (as in Lab session 1). Do this analysis sampling the series first **weekly** and then **monthly** periods. For computing causality among two pairs of assets A and B , first estimate the optimal lag l in the range of $[1, 5]$, using `VAR.select_order` with AIC (Akaike Information Criterion); and after estimate causality for A and B at lag l .

Tabulate **and comment** your results (four tables, one for returns, other for volatility and each for weekly and for monthly sampling), so that *cause* $\rightarrow$ *effect* goes from row to column,

and each entry the value of l if there is causality, or 0 otherwise. Example

	SPY	XLK	XLFX	...
SPY	-	3	0	...
XLK	0	-	2	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

so, first entry means $SPY \rightarrow XLK$ at lag 3 (optimal) Causality should be considered significant at the 5% level. Comment your results. (Some hints for your narrative: Among which assets there is true causality, contemporaneous correlation? Are these causalities related to economic events? Show plots of network of causalities. Returns network: Who drives the price? Volatility network: Where does the "fear" spill over?)

2. (40%) **Neural Networks vs. LSTM vs. Gaussian process. Forecasting with fundamental indicators.** Predict the SP500 with the financial indicators assigned to your team in the google spreadsheet (ep, dp, de, dy, dfy, bm, svar, ntis, infl, tbl (see R script Helper or Python script helper), some lagged series of these indicators and lags of the target using a Neural Network, a LSTM and a GP regression with your desired kernel. Predict return, or price, or direction (up or down). For which target works best? Select appropriate lags and do feature selection (e.g., using causality, (distance) correlation, VAR-test, Lasso, or any other criterion). Fine tune your models, but careful not to overfit: for NNet try hidden layer size (width) or depth, activation function; GP, try different kernels (stationary and non-stationary) from library or define your own; LSTM try units, timestep, layers. Do a sensitivity analysis on the exogenous variables. Are all good for prediction? (The dataset is `goyal-welch2022Monthly.csv` and work within the period 1927/2021. The script `PyLab5_MLmodels.ipynb` gives you a basic set-up, which needs to be improved. `PyLab5_GausProc.ipynb` shows how to define your own kernel.)
3. (20%) Compare forecasting modeling in previous task against $ARMA(p,q) + GARCH(1,1)$ fit (as baseline model) together with a normalization transformation of variables.¹ (You may use R package `bestNormalize` to automatically find and apply the best normalizing transformation and invert results to original domain -for the case of predicting return or price, or implement your own normalization transformation from the ones defined in Lecture or search for others.)

¹Note that log returns of prices is considered a normalizing transformation. Explore other normalizing transformations.

4. (5%) Write a summary of main lessons learnt from Lectures 1-5 (e.g. 2-3 keypoints for each lecture).
5. Write each authors contribution. E.g. AA: worked the proof in Prob 1; BB wrote the report; AA & BB clean data for prob 2, 3, BB wrote code and set experiments, AA made analysis and wrote report; or “All authors contributed equally”.